

# Xijia “Polina” Zhang

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## Research Interests

I am deeply passionate about **mechanistic interpretability**. I aim to reverse-engineer foundation models to be steerable, grounded, and transparent, and to bridge their latent representations with human-level semantics through language.

## Education

**Georgia Institute of Technology**

Aug. 2024 – Dec. 2029

*Ph.D. in Robotics*

**University of Michigan**

Sep. 2022 – May. 2024

*B.S.E in Computer Science*

**Shanghai Jiao Tong University**

Sep. 2020 – Aug. 2024

*B.E in Electrical and Computer Engineering*

## Publications

[“Model-Agnostic Policy Explanations with Large Language Models”](#) [↗](#)

**Zhang Xi-Jia**, Yue Guo, Shufei Chen, Simon Stepputtis, Matthew Gombolay, Katia Sycara, Joseph Campbell  
COLM 2025 (**top 5%**)

- Propose a method for generating natural language explanations of agent behavior based only on observed states and actions, without access to the agent’s underlying model.

[“Towards Human-Free Semantic Interpretability in Reinforcement Learning via Vision-Language](#)

[Models”](#) [↗](#)

(in submission)

Zhaoxin Li\*, **Zhang Xi-Jia**\*, Batuhan Altundas, Letian Chen, Rohan Paleja, Matthew Gombolay

- Develop an automated RL framework that leverages foundational models for semantic feature extraction and interpretable tree-based models for policy optimization.

[“Thinking Deeper Without Growing Larger: Scaling Virtual Depth in Pretrained LLMs for Enhanced](#)

[Reasoning”](#)

(in submission)

Shivang Chopra, **Zhang Xi-Jia**, Chengyue Huang, Karmesh Yadav, Yusuf Ali, Zsolt Kira

- Introduce dynamic layer routing via lightweight adapters that learn to adaptively skip or loop transformer blocks. Analyze activations, entropy, and model responses for mechanistic insights.

[“Learning Effective Action Advising in the Face of Changing Rewards”](#) [↗](#)

Yue Guo, **Zhang Xi-Jia**, Simon Stepputtis, Joseph Campbell, Katia Sycara

CoLLAs 2024 (**Oral**)

- Enable the teacher policy to continually learn and adapt its reward function through ongoing observation of the student when providing action advices.

[“Sensor Array Optimization for the Electronic Nose via Different Deep Learning Methods”](#) [↗](#)

**Zhang Xi-Jia**\*, Tao Wang\*, Wangze Ni, Yongwei Zhang, Wen Lv, Min Zeng, Jianhua Yang, Nantao Hu, Rui Zhan, Guang Li, Zhiqiang Hong, Zhi Yang  
Sensors and Actuators: B, 2024

- Compare lightweight machine learning models on an Electronic Nose and investigate how their performance scales with dataset size.

“Teaching the Teacher: Enhancing Human-to-Robot Skill Demonstration with Live Foundation Model and Augmented Reality Feedback”

Nina Moorman, Matthew Luebbers, **Zhang Xi-Jia**, Zulfiqar Zaidi, Marcus Lau, Yixing Yao, Megan Langwasser, Letian Chen, Sanne van Waveren, Matthew Gombolay (in submission)

- Support non-expert users in providing kinesthetic demonstrations through language explanations and policy visualizations.
- Contributions: Language explanations.

“Communication and Verification in LLM Agents towards Collaboration under Information Asymmetry

Run Peng, Ziqiao Ma, Amy Pang, Sikai Li, **Zhang Xi-Jia**, Yingzhuo Yu, Cristian-Paul Bara, Joyce Chai (in submission)

- Investigate the performance of LLM agents in human-robot collaboration tasks under information asymmetry.
- Contributions: TIAGo robot development.

Skills

|                        |                                                |
|------------------------|------------------------------------------------|
| Toolchain              | ROS, ROS2, Linux, Docker, Huggingface, RLlib   |
| Frameworks & Libraries | Gym, NLTK, Transformers, Pytorch, Scikit-Learn |
| Foundation Model       | LLM fine-tuning, prompting; LoRA, PPO, GRPO    |
| Robotics Development   | TIAGo, Jaco Arm, Khepera, Fetch; RViz, Gazebo  |

Honors & Awards

|           |                                             |                                 |
|-----------|---------------------------------------------|---------------------------------|
| Sep. 2024 | Robotics Fellowship                         | Georgia Institute of Technology |
| Mar. 2024 | James B. Angell Scholar                     | University of Michigan          |
| Apr. 2024 | Elected Member                              | Tau Beta Pi, Michigan Gamma     |
| Dec. 2023 | Dean’s List                                 | University of Michigan          |
| Apr. 2023 | Dean’s List                                 | University of Michigan          |
| Dec. 2022 | Dean’s List                                 | University of Michigan          |
| May. 2023 | Chun-Tsung Scholar                          | Shanghai Jiao Tong University   |
| Oct. 2021 | Rongchang Innovation Scholarship Nomination | Shanghai Jiao Tong University   |
| Nov. 2021 | Silver Medal                                | The University Physics Contest  |